Lab Exercise 1- Docker Fundamental Commands

Objective

Learn and practice the fundamental Docker CLI commands to:

- Manage images and containers
- Understand container lifecycle
- Build and run applications using Docker

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1. Setup

Prerequisites

- Docker installed on your system

  (Check using: docker --version)
- Internet access to pull images

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2. Basic Docker Commands

Step 1: Verify Installation

```
docker –version
```

```
Microsoft Windows [Version 10.0.26200.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NK>docker pull ubuntu:latest
latest: Pulling from library/ubuntu
a3629ac5b9f4: Pull complete
1baf05536e37: Download complete
Digest: sha256:cd1dba651b3080c3686ecf4e3c4220f026b521fb76978881737d24f200828b2b
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest

C:\Users\NK>docker images
                                                                                    i Info →   U  In Use
IMAGE           ID             DISK USAGE   CONTENT SIZE   EXTRA
ubuntu:latest   cd1dba651b30        119MB         31.7MB

C:\Users\NK>
```

```
docker info
```

Expected Output: Docker version and system details.

Step 2: Pull an Image from Docker Hub

```
docker pull ubuntu:latest
```

Explanation: Downloads the latest Ubuntu image from Docker Hub.

Check available images:

```
docker images
```

```
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C:\Users\NK>docker pull ubuntu:latest
latest: Pulling from library/ubuntu
a3629ac5b9f4: Pull complete
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Digest: sha256:cd1dba651b3080c3686ecf4e3c4220f026b521fb76978881737d24f200828b2b
Status: Downloaded newer image for ubuntu:latest
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C:\Users\NK>docker images
                                                                    i Info →   U  In Use
IMAGE           ID             DISK USAGE   CONTENT SIZE   EXTRA
ubuntu:latest   cd1dba651b30        119MB         31.7MB

C:\Users\NK>
```

Step 3: Run a Container

Run Ubuntu interactively:

```
docker run -it --name myubuntu ubuntu bash
```

Now you're inside a running container (prompt will change).

Exit:

```
Exit
```

```
Microsoft Windows [Version 10.0.26200.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NK>docker run -it --name myubuntu ubuntu bash
root@b5d54eb41531:/#
root@b5d54eb41531:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@b5d54eb41531:/#
```

Step 4: List Containers

- Show running containers:

```
docker ps
```

- Show all containers (including stopped):

```
docker ps –a
```

```
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(c) Microsoft Corporation. All rights reserved.

C:\Users\NK>docker run -it --name myubuntu ubuntu bash
root@b5d54eb41531:/#
root@b5d54eb41531:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@b5d54eb41531:/# exit
exit

C:\Users\NK>
```

Step 5: Start / Stop / Remove Containers

```
# Start a stopped container

docker start <container_id>

# Stop a running container

docker stop <container_id>

# Remove a container

docker rm <container_id>
```

```
C:\Users\NK>docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED              STATUS                      PORTS     NAMES
b5d54eb41531   ubuntu    "bash"    About a minute ago   Exited (0) 33 seconds ago             myubuntu

C:\Users\NK>
```

Tip: Use docker ps -a to get container IDs.

Step 6: Remove Images

docker rmi ubuntu:latest

```
C:\Users\NK>docker start myubuntu
myubuntu

C:\Users\NK>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED              STATUS         PORTS     NAMES
b5d54eb41531   ubuntu    "bash"    About a minute ago   Up 7 seconds             myubuntu

C:\Users\NK>docker stop myubuntu
myubuntu

C:\Users\NK>
```

```
C:\Users\NK>docker rm myubuntu
myubuntu

C:\Users\NK>docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\NK>
```

Note: You must stop and remove all containers using that image first.

```
C:\Users\NK>docker rmi ubuntu:latest
Untagged: ubuntu:latest
Deleted: sha256:cd1dba651b3080c3686ecf4e3c4220f026b521fb76978881737d24f200828b2b

C:\Users\NK>docker images
                                                                                  i Info →   U In Use
IMAGE   ID            DISK USAGE   CONTENT SIZE   EXTRA

C:\Users\NK>
```